

CAD-SEL: A dual-modal colonoscopy dataset of subepithelial lesion

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Data Descriptor**Title**

CAD-SEL: A dual-modal colonoscopy dataset of subepithelial lesion

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Abstract

Subepithelial lesions (SELs) of the gastrointestinal tract encompass a heterogeneous spectrum of histology, ranging from benign to malignant. Their detection during colonoscopy is critical yet challenging, as endoscopic interpretation is highly operator-dependent with variable diagnostic performance. Artificial intelligence (AI) offers great promise for assisting endoscopists in SEL detection and assessing, yet progress has been limited by the lack of large, diverse, and well-annotated datasets. To address this, we present CAD-SEL dataset, the first publicly available dual-modality colonoscopy dataset for SEL detection. The dataset comprises 4,912 images from 641 patients, collected across six medical centers in China, with paired white light endoscopy and endoscopic ultrasound images. Expert endoscopists provided histology-based multi-class annotations and clinically relevant binary classifications (potentially malignant versus benign) ensuring both research flexibility and AI-driven applicability. Initial technical validation with several state-of-the-art object detection models demonstrates the dataset's potential to support AI-based computer-assisted detection systems. CAD-SEL dataset provides a high-quality open-access benchmark to advance AI-assisted SEL analysis, support reproducibility, and facilitate broader clinical translation.

Background & Summary

Subepithelial lesions (SELs), historically termed submucosal tumors, are masses, bulges, or impressions of the gastrointestinal (GI) tract covered by normal-appearing epithelium.¹ Unlike intraepithelial neoplasia, which originates from the mucosal layer, SELs constitute another major category of GI lesions, arising from nonmucosal layers including the muscularis mucosa, submucosa, or muscularis propria.² They are often asymptomatic and represent a heterogeneous spectrum of histology, ranging from benign to malignant entities.³ While a majority of SELs are considered benign, approximately 13%-23% demonstrate precancerous or malignant potential, depending on anatomical location, size and histological subtype⁴, having potential for malignant transformation and metastatic spread.^{5,6} Consequently, timely and accurate diagnosis is essential for determining appropriate therapeutic strategies and optimizing patient outcomes.

Current clinical practice relies on multi-modal colonoscopy for the detection of colorectal SELs.⁷ Conventional white-light endoscopy (WLE) provides surface visualization of the mucosa, enabling assessment of features such as lesion size, shape, mobility and color.^{8,9} Endoscopic ultrasound (EUS) complements WLE by ascertaining the exact size, depth of origin and additional morphologic features which aid differential diagnosis.^{10,11} While these modalities are indispensable for accurate and early

detection of SELs, their interpretation remains highly operator-dependent owing to diverse histologic subtypes and overlapping endoscopic appearances, leading to considerable variability in diagnostic accuracy across endoscopists.¹²

Artificial intelligence (AI) has shown great potential in advancing medical image interpretation, with significant progress in gastroenterology for tasks such as lesion detection, classification, and segmentation.^{13–15} The development of reliable AI systems relies on large, diverse, and well-annotated datasets reflecting the heterogeneity of real-world clinical practice. Although several studies have compiled SELs datasets for research purposes^{14,16–23}, most remain proprietary and are not publicly accessible. At present, no openly available public database dedicated to SELs exists, posing a barrier to reproducibility, benchmarking, and broader application of AI-driven analysis.

To contextualize the contribution of CAD-SEL, we systematically compared it with existing SEL-related datasets reported in the literature (Table 1). As summarized in Table 1, prior studies have compiled SEL datasets ranging from 60 to 1,366 patients, predominantly focusing on the upper gastrointestinal tract — particularly the stomach — for differentiating GISTs from leiomyomas or other mesenchymal tumors.¹⁷ collected 10,439 EUS images from 752 patients across four centers, and Hirai et al.¹⁸ assembled the largest dataset of 16,110 EUS images from 631 patients across 12 hospitals, both targeting upper gastrointestinal SELs with up to five histological subtypes. However, all existing datasets remain proprietary and are not publicly accessible. Moreover, the majority rely solely on EUS as the imaging modality, with only Zhu et al.²² incorporating both WLE and EUS, albeit limited to gastric lesions and without public access. Notably, no dataset has been specifically designed for colorectal SELs, despite the distinct epidemiological and histopathological characteristics of SELs at this anatomical location. CAD-SEL addresses these gaps as the first publicly available, multicenter, dual-modality (WLE and EUS) dataset dedicated to colorectal SELs, encompassing 4,912 annotated images from 641 patients with five histologically confirmed subtypes and bounding-box annotations for object detection tasks.

In this study, we introduce CAD-SEL, the first publicly available dual-modality colonoscopy dataset for SELs. The resource includes paired WLE and EUS images of histologically confirmed colorectal SELs, collected across multiple centers and annotated by expert endoscopists. To demonstrate its utility, we performed baseline experiments with state-of-the-art AI models for lesion detection under single- and multi-task learning frameworks. The general purpose of this diverse dataset is to advance research into AI-assisted SELs recognition, foster reproducibility, and support more accurate and reliable clinical decision-making in colonoscopy.

Study	Year	Public	Anatomical Site	Modality	Centers	Patients	Images	Subtypes	Annotation
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Oh et al. ¹⁴	2021	No	Gastric	EUS	1	168	546	2	Cls+Det
Yang et al. ¹⁷	2022	No	Upper GI	EUS	4	752	10,439	2	Cls
Hirai et al. ¹⁸	2022	No	Upper GI	EUS	12	631	16,110	5	Cls
Kim et al. ¹⁹	2020	No	Gastric	EUS	1	—	1,117	3	Cls
Seven et al. ²⁰	2022	No	Upper GI	EUS	1	145	1,362	2	Cls
Minoda et al. ²¹	2020	No	Gastric	EUS	1	60	—	2	Cls
Zhu et al. ²²	2022	No	Gastric	WLE+EUS	2	1,366	—	3	Cls
Dong et al. ²³	2024	No	Upper GI	EUS	5	1,342	5,419	2	Cls+Det
He et al. ²⁴	2023	Planned	Upper GI	EUS	1	514	—	2	Cls+Det
CAD-SEL (Ours)	2025	Yes	Colorectal	WLE+EUS	6	641	4,912	5	Cls+Det

Table 1. Comparison of CAD-SEL with existing SEL-related datasets. Cls = classification labels; Det = detection with bounding-box annotations; Upper GI = upper gastrointestinal tract (esophagus, stomach, duodenum); GIST = gastrointestinal stromal tumor; NET = neuroendocrine tumor; "—" indicates data not explicitly reported.

Methods

Data cohorts.

The dual-modal colonoscopy dataset (CAD-SEL) comprises paired WLE and EUS images of colorectal SELs, collected from six medical centers in China, including the First Affiliated Hospital of Zhejiang University (Center 1), Ningbo First Hospital (Center 2), Taizhou Hospital (Center 3), Huzhou Central Hospital (Center 4), Shengzhou People's Hospital (Center 5), and Sanmen County People's Hospital (Center 6). Patients with histologically confirmed colorectal SELs were included, while cases with poor-quality images or without histological confirmation were excluded. The following criteria were applied

for image exclusion: (1) with poor-quality image datasets due to issues like defocus, inadequate inflation, and blurring; (2) Excessive specular reflection (glare) obscuring more than 50% of the lesion area; (3) Food residue, fecal matter, or inadequate bowel preparation preventing clear visualization of the lesion; (4) Images with significant artifacts (e.g., water jet, instrument interference) that obscured the region of interest. The patient and image selection process is illustrated in a PRISMA-style flowchart (Fig 1), including the number of cases screened, excluded (with reasons), and finally included at each stage. The included histological subtypes — NET, lipoma, leiomyoma, and non-neoplastic lesions — were selected based on their clinical prevalence among colorectal SELs. In the colorectal region, these four categories account for the vast majority of histologically confirmed SELs.¹⁶ Other neoplastic subtypes, such as gastrointestinal stromal tumors (GISTs), malignant lymphomas, and metastatic tumors, are far more prevalent in the upper gastrointestinal tract (particularly the esophagus and stomach) but are exceedingly rare in the colorectum. During our multicenter data collection, the number of these rare cases was insufficient to form statistically meaningful categories for AI model training and were therefore not included in the current version of the dataset.

Patient data were retrospectively collected from January 2019 to February 2024. Patients aged over 18 years who underwent colonoscopy and were diagnosed with colorectal SELs (NETs, lipomas, leiomyomas, and non-neoplastic lesions including inflammatory hyperplasia) were included. Image acquisition was performed using standard colonoscopes (EC-720R/M, Fujifilm Medical Systems, Tokyo, Japan; CF-H290I and CF-HQ290I, Olympus Medical Systems, Tokyo, Japan), with endoscopic ultrasound performed using an ultrasound processor (EU-ME2 PREMIER PLUS, Olympus Medical Systems, Tokyo, Japan) equipped with a 12-MHz miniprobe (UM-2R, Olympus Corporation, Tokyo, Japan). For each patient, WLE and EUS examinations were performed either during the same endoscopic session or within a maximum interval of three days, with clinical records confirming no interval change in lesion characteristics. The pairing of WLE and EUS images was established at the patient level based on the confirmed anatomical location of the same lesion. In addition, WLE and EUS images were aligned through a two-step process. First, both modalities were acquired during the same endoscopic session using an integrated system with a picture-in-picture (PIP) display, allowing simultaneous visualization of white-light and ultrasound images to ensure acquisition of the same lesion at the same anatomical location. Second, cross-modality correspondence was independently verified by two expert reviewers during annotation based on lesion location and clinical records. As all participating centers employed the same equipment models, specifications are reported collectively. The per-center distribution of patients and images is provided in Supplementary Table S1. The Clinical Research Ethics Committee of the First Affiliated Hospital of Zhejiang University reviewed the study protocol and granted ethical approval

(Approval No. 2024-0129), including a waiver of informed consent, with agreements from all participating centers.

Anonymization protocol.

To protect patient privacy and comply with data protection regulations, all data were anonymized prior to inclusion in the dataset. Patient identifiers were removed and replaced with unique pseudonyms consisting of a categorical prefix and a numeric index (e.g., Lipoma-1, Leiomyoma-1) allowing consistent reference to images from the same case while preventing re-identification. This pseudonymization method ensures that no directly identifiable personal information is retained.

For image data, all visible text overlays (e.g., patient names, dates, institutional identifiers) were manually cropped or digitally removed, and only the endoscopic visual content was retained. This step was conducted to prevent indirect identification through burned-in metadata or visible identifiers that appear on the images themselves. The anonymization protocol was applied uniformly across all WLE and EUS images, ensuring that no protected information could be inferred from either pixel data or metadata.

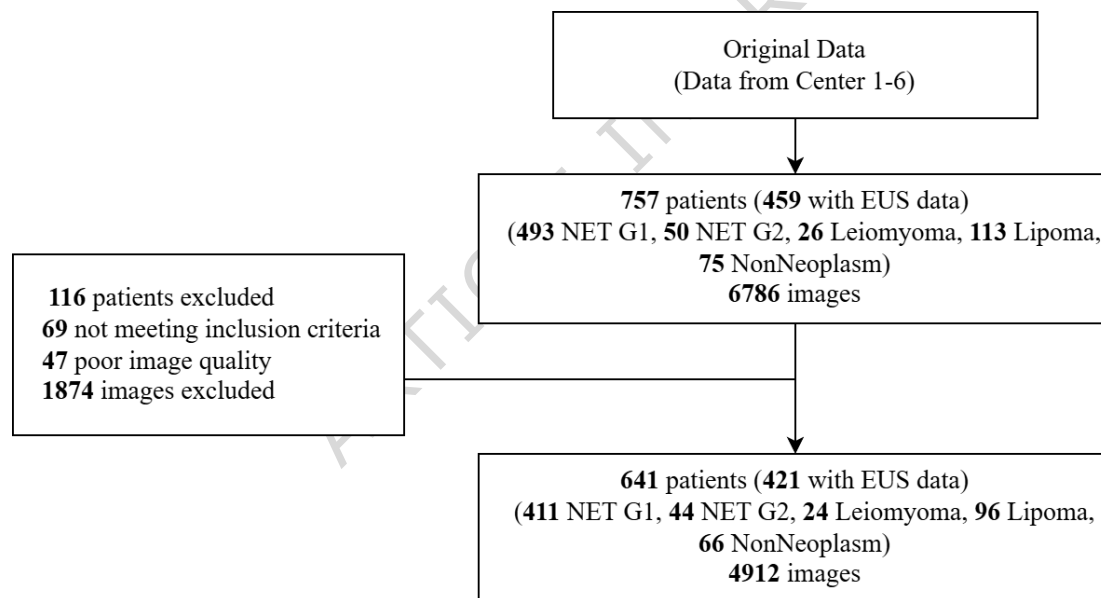


Fig 1. Flowchart of patient and image selection for the CAD-SEL dataset.

Annotation strategies and quality assurance.

First, histology-based multi-class labels were assigned exclusively based on pathological diagnoses confirmed through endoscopic resection (e.g., endoscopic mucosal resection, endoscopic submucosal dissection). These included four subtypes of SELs according to the World Health Organization (WHO) classification: neuroendocrine tumor (NET), leiomyoma, lipoma, and non-neoplastic lesion (NonNeoplasm). NETs were further sub-classified into Grade 1 (G1) and Grade 2 (G2).²⁵ Second, a clinically relevant binary label was applied to distinguish potentially malignant from benign lesions. Specifically, NETs were considered malignant or potentially malignant, while lipomas, leiomyomas, and

NonNeoplasm were considered benign.^{7,26} This labeling scheme supports both histology-based multi-class and clinically relevant binary classification tasks.

In the following data annotation phase, a team of 4 expert endoscopists, each with more than 10 years of endoscopic experience were involved, using an online annotation tool called Label Studio platform (<https://labelstud.io>). Two expert endoscopists independently performed the initial annotation. They delineated regions of interest (ROIs) corresponding to the lesions by drawing bounding boxes around visible SELs in both WLE and EUS images. To ensure annotation accuracy, a separate panel of two experts reviewed all initial annotations, including verifying the lesion localization, consistency with histological findings, and cross-modality coherence. Any discrepancies were resolved through consensus discussions involving all four endoscopists.

This multi-stage annotation and verification pipeline ensured accuracy, consistency and reproducibility of the annotations. The resulting bounding box labels provide high-confidence lesion localization, supporting robust training and evaluation of AI models for lesion detection and classification tasks.

Data Records

The complete CAD-SEL dataset is publicly available via figshare.²⁷ In total, it contains 4,912 labelled images from 641 patients with colorectal SELs, stored in TIFF format with RGB color depth (24-bit). The original images were acquired in TIFF or PNG format depending on the endoscopic system at each center; all images were converted to TIFF with lossless compression to ensure format consistency. Image resolutions range from 362×290 to 1920×1080 pixels (median: 1160×1008). No resizing, resampling, color normalization, or contrast enhancement was applied. For a subset of images, non-informative black borders resulting from the endoscopic field of view were cropped to retain only clinically relevant content. Apart from this border removal and the de-identification process described in the Methods section, no other preprocessing was performed. All cases are categorized into five SELs subtypes: NET_G1, NET_G2, leiomyoma, lipoma, and NonNeoplasm. For simplicity, the term NET is used collectively to refer to both NET_G1 and NET_G2 throughout this article. Representative examples from each lesion subtype are shown in Fig 2. and an overview of the data records is provided in Table 2.

The dataset is packaged as a compressed ZIP file containing two main folders (*Images* and *Labels*), as well as a metadata file (*metadata.xlsx*). Within each folder, subdirectories labelled Center-1 through Center-6 correspond to the six contributing institutions. Each center folder is further organized by colonoscopy modality into WLE-Set and EUS-Set, and with each modality folder, images are grouped into five subcategories named NET_G1, NET_G2, Leiomyoma, Lipoma, and NonNeoplasm. Patient-level subfolders contain multiple images for the same case, ensuring consistency between WLE and EUS modalities.

As detailed in Table 3, the accompanying metadata file provides patient-level descriptors. Columns of the file include Patient Index, Center, Category, EUS performed (1 = performed, 0 = not performed), Patient ID, Age (years), Sex (0 = female, 1 = male), and Lesion Location. All patients have corresponding WLE images, and an EUS performed value of 1 indicates the availability of paired EUS images.

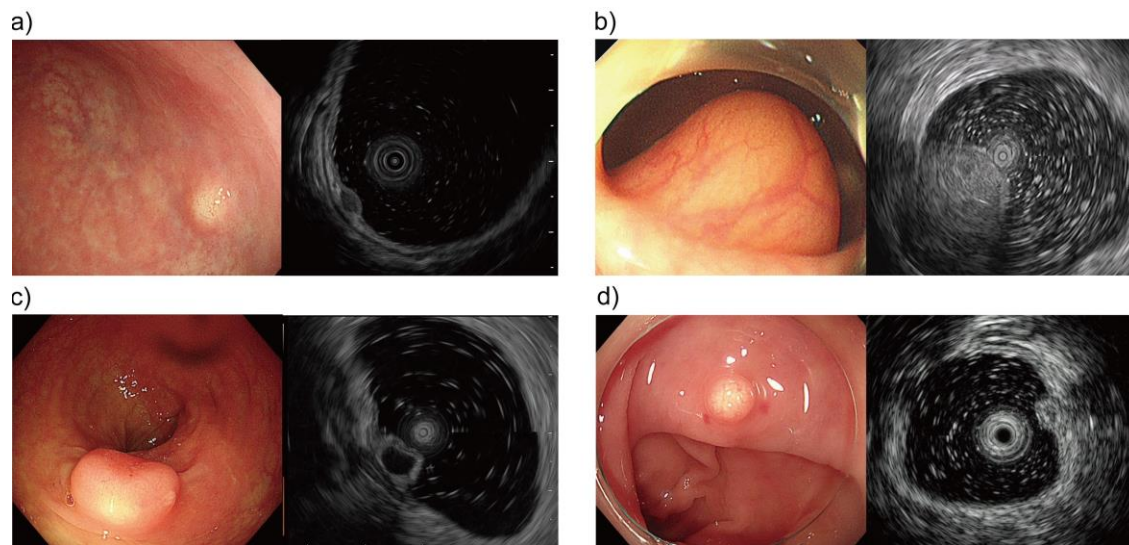


Fig 2. Multi-modal image examples of the labelled classes in the CAD-SEL dataset. Figures were taken from: a) Neuroendocrine Tumor (NET) dataset; b) Lipoma dataset; c) Leiomyoma dataset; d) NonNeoplasm dataset.

Category	Number of patients	Number of images
WLE set	641	2,817
NET	455	1,945
NET_G1	411	1,749
NET_G2	44	196
Lipoma	96	453
Leiomyoma	24	137
NonNeoplasm	66	282
EUS set	421	2,095
NET	325	1,602
NET_G1	299	1,444
NET_G2	26	158
Lipoma	49	254

Leiomyoma	14	79
NonNeoplasm	33	160

Table 2. Overview of the data records in CAD-SEL dataset

Index	Center	Category	EUS performed	Name	Age (y)	Sex (Female=0, Male=1)	Size (cm)	Location
0	Center-1	NET_G1	1	NET_G1-0	62	1	1.1	rectum
558	Center-2	Leiomyoma	1	Leiomyoma-18	56	0	0.7	descending colon
629	Center-3	NonNeoplasm	1	NonNeoplasm-65	64	1	0.8	ileocecal region
633	Center-4	Lipoma	1	Lipoma-95	57	0	1.0	transverse colon
640	Center-6	NET_G2	1	NET_G2-43	59	1	1.0	rectum

Table 3. Samples from the metadata.xlsx file.

Technical Validation

Dataset characteristics.

This multicenter CAD-SEL dataset comprises 2,817 WLE images of 641 patients and 2,095 EUS images of 421 patients with colorectal SELs across 6 medical institutions. The mean age of the cohorts was 52.0 years, with 68% of the patients between 30 and 60 years. The sex distribution was balanced (51% male, 49% female). Most lesions were located in the rectum (81%), followed by the sigmoid colon (5%), descending colon (3%), transverse colon (3%), ascending colon (5%), and ileocecal region (3%). In terms of size, over half of the colorectal SELs (54%) measured 0.5-1 cm, 20% were <0.5cm, 20% were 1-2cm, and 6% exceeded 2cm. A detailed distribution of clinical data is provided in Fig 3.

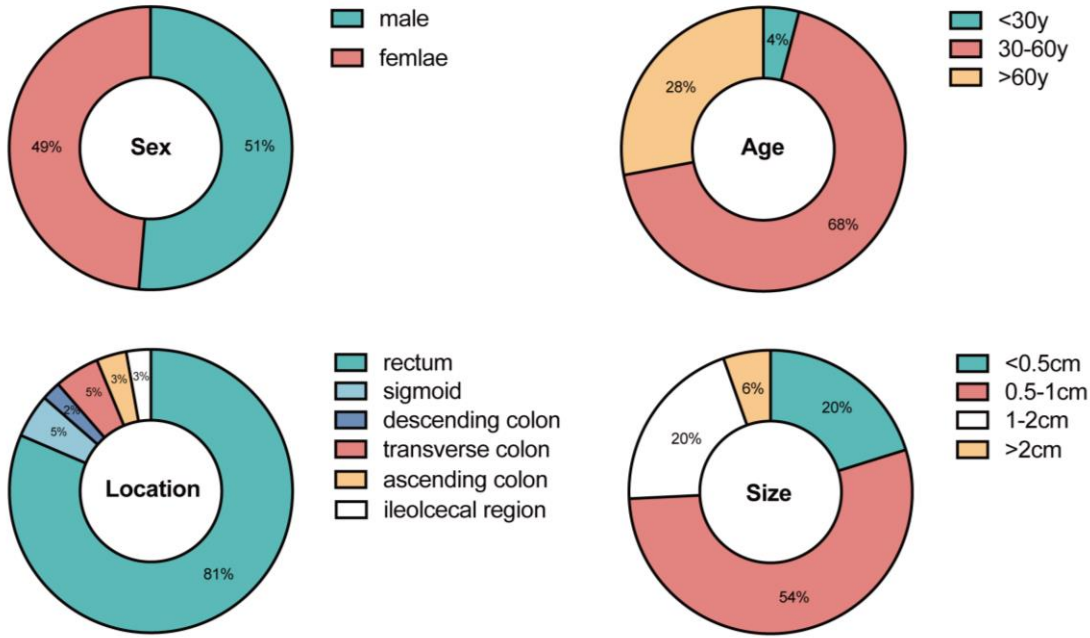


Fig 3. Clinical Data Distribution. This figure presents pie charts depicting the distribution of sex, age, location and size within the CAD-SEL dataset.

Data validation.

This section presents a technical validation to assess the quality and utility of CAD-SEL for automated SELs detection. Four state-of-the-art object detection models were trained and evaluated:

Fasterrcnn_Resnet, Fasterrcnn_Resnet50_FPN_V2, Fasterrcnn_Mobilenet and

Fasterrcnn_Mobilenet_320.²⁸⁻³⁰ Experiments were conducted under two labeling schemes: (i) multi-class detection, distinguishing between all SELs categories, and (ii) binary detection, differentiating NET from non-NET. This dual-task setup reflects both detailed histology-based classification and clinically relevant (potentially malignant versus benign) classification.

All models were optimized with classification loss and bounding-box regression loss. Class weight balancing was further implemented by assigning weights inversely proportional to the frequency of each category to address dataset imbalance. Images from the same patient were assigned exclusively to either the training or validation set within each fold. Representative predictions from the best performing model are shown in Fig 4.

We compute standard metrics used for assessing detection performances that include Average Precision at an Intersection over Union threshold of 0.5 (AP@0.5), recall, precision, and F1 score (see Table 4 and Table 5). These metrics are defined as follows:

AP@0.5, a widely used metric in object detection, measures the area under the precision–recall curve. In this context, a predicted bounding box is considered as correct if it overlaps with a ground truth box by at least 50% IoU and the class label matches.

Recall, also referred to as the true positive rate (TPR), measures the proportion of actual positive samples that are correctly identified by the model.

$$recall = \frac{TP}{TP + FN}$$

Precision, also known as the positive predictive value (PPV), measures the proportion of predicted positive samples that are true positives.

$$precision = \frac{TP}{TP + FP}$$

F1 score, is the harmonic mean of precision and recall, offering a more comprehensive evaluation of the model's performance. It provides a single summary measure that is especially informative in datasets with class imbalance.

$$F1\ score = 2 \times \frac{precision \times recall}{precision + recall} = \frac{2TP}{2TP + FP + FN}$$

The multi-class detection results are summarized in Table 4, reporting the overall performance of four models in recognizing SELs into histology-based categories using both WLE and EUS images. On WLE data, the models achieved AP@0.5 between 0.8593 and 0.8817, with F1-scores ranging from 0.7816 to 0.8790. EUS-based detection showed greater variability and reduced performance compared to WLE modality, with F1-scores varying from 0.6890 to 0.7661. Among the tested models, Fasterrcnn_Mobilenet demonstrated the highest performance, with an AP@0.5 of 0.8817 and F1-score of 0.8508 on WLE-based detection.

Binary detection results are presented in Table 5, with a focus on identification of NET versus non-NET. When identifying NET, models showed stable performance, with AP@0.5 ranging from 0.8851 to 0.9119 on WLE images and AP@0.5 ranging from 0.7257 to 0.8143 on EUS images. For non-NET, performance remained strong on WLE data (AP@0.5: 0.8068-0.8784) but was markedly lower on EUS data (AP@0.5: 0.3383-0.4213). Among the tested models, Fasterrcnn_Resnet achieved the highest performance, with AP@0.5 of 0.9119 on WLE and 0.8011 on EUS for NET detection. Overall, these results indicate that WLE modality provides relatively stronger features for both multi-class and binary SELs detection tasks.

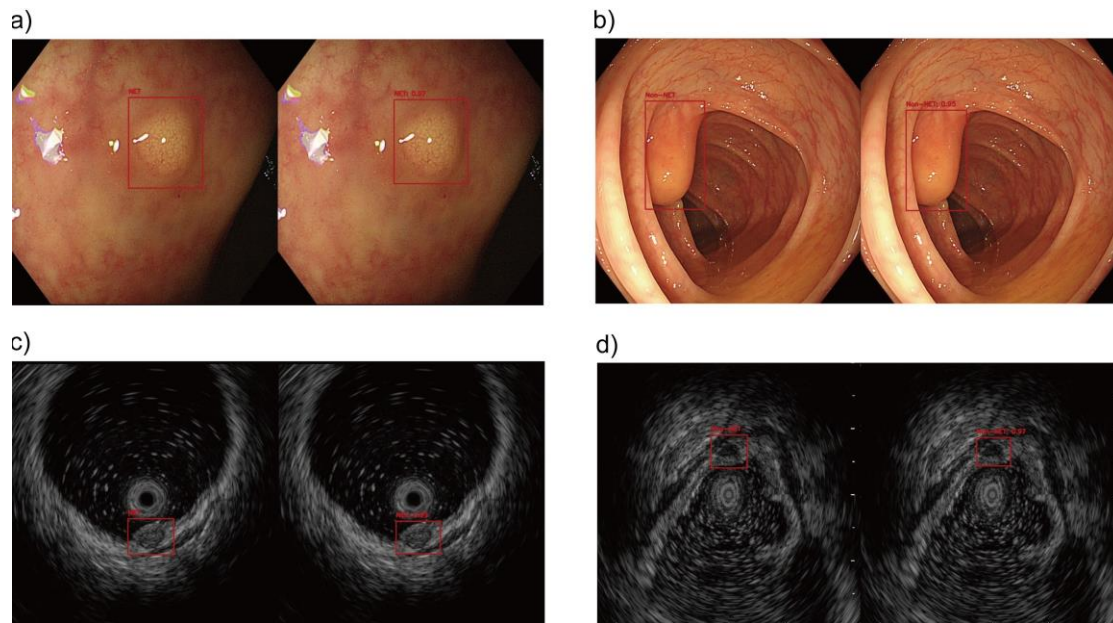


Fig 1. Sample images from the testing dataset, with predictions from the best-performing model. The left column displays ground truth annotations (red boxes), while the right column shows corresponding model predictions (red boxes with probability scores). Examples of binary classification are provided: (a) prediction of NET based on white-light endoscopy (WLE) imaging with a probability of 0.97; (b) prediction of Non-NET based on WLE with a probability of 0.95; (c) prediction of NET from endoscopic ultrasound (EUS) imaging with a probability of 0.99; (d) prediction of Non-NET from EUS imaging with a probability of 0.97.

Tested Models	Colonoscopy Modality	AP@0.5	Precision	Recall	F1-score
Fasterrcnn_Resnet	WLE	0.8593	0.6824	0.9146	0.7816
	EUS	0.6112	0.5938	0.8206	0.6890
Fasterrcnn_Resnet50_FPN_V2	WLE	0.8792	0.6772	0.929	0.7834
	EUS	0.5769	0.6071	0.8129	0.6951
Fasterrcnn_Mobilenet	WLE	0.8817	0.8014	0.9067	0.8508
	EUS	0.5768	0.7764	0.7561	0.7661
Fasterrcnn_Mobilenet_320	WLE	0.8641	0.8661	0.8922	0.8790
	EUS	0.5320	0.7730	0.6840	0.7258

Table 4. Overall performance metrics in multi-category detection tasks of the models.

Tested Models	Colonoscopy	AP@0.5 (NET)	AP@0.5 (non-NET)
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	modality		
Fasterrcnn_Resnet	WLE	0.9119	0.8068
	EUS	0.8011	0.4213
Fasterrcnn_Resnet50_FPN_V2	WLE	0.9105	0.8480
	EUS	0.7814	0.3725
Fasterrcnn_Mobilenet	WLE	0.8851	0.8784
	EUS	0.8143	0.3393
Fasterrcnn_Mobilenet_320	WLE	0.9023	0.8259
	EUS	0.7257	0.3383

Table 5. Binary detection results of the models for NET versus non-NET, reported as AP@0.5.

Usage Notes

To the best of our knowledge, CAD-SEL is the first publicly available dual-modal colonoscopy dataset for computer-aided subepithelial lesion detection, including paired WLE and EUS images. CAD-SEL is intended to facilitate the development and benchmarking of machine learning algorithms for accurate detection and assessment of SELs in the GI tract. The inclusion of expert annotation, histology-based and clinically relevant labels ensures a high-quality resource that supports both research and clinical training applications.

Users should be aware that class imbalance, particularly the limited representation of certain lesion subtypes (e.g. leiomyoma), may affect model performance. To address this, strategies such as class-weight balancing³¹ or advanced sampling methods are recommended to improve detection accuracy for rare but clinically important SEL subtypes. Moreover, although CAD-SEL is a multicenter and dual-modality dataset, the overall sample size remains relatively moderate, which may limit model robustness. Future studies will require larger-scale SEL datasets to better support the real-world application of deep learning models. Furthermore, although all participating centers employed endoscopic systems from similar manufacturers and model series, device-related differences were not systematically analyzed in this study and may still introduce variability. Future work should explicitly consider and evaluate inter-device variability to further assess model generalizability.

Additionally, it is also essential to perform dataset partitioning at the patient level, rather than the image level, to avoid data leakage from correlated images appearing in both training and testing sets. Adhering to these precautions helps maintain the integrity of model evaluation and ensures more reliable generalization. Moreover, the current dataset does not include rare colorectal SEL subtypes such as GISTs, malignant lymphomas, or metastatic tumors, owing to their extremely low incidence in the

colorectal region¹⁶. During our multicenter data collection, the number of these rare cases was insufficient to form statistically meaningful categories for AI model training and were therefore not included in the current version of the dataset. Future iterations of the dataset aim to incorporate these rare subtypes as additional cases become available across participating centers.

For transparency and reproducibility, example scripts demonstrating patient-level dataset partitioning and weighted training procedures are provided in the project's GitHub repository, as detailed in the Code Availability section. We encourage researchers to use CAD-SEL for comparative analyses and to contribute enhancements or novel annotations to foster community-driven advancement. Furthermore, the results encourage further research to optimize detection algorithms tailored to each modality's unique strengths within the CAD-SEL dataset. CAD-SEL is distributed under the Creative Commons Attribution 4.0 International (CC BY 4.0) License, permitting use, sharing, adaptation, and redistribution in any medium, provided appropriate credit is attributed to the original authors and source.

Code Availability

In addition to releasing the data, we also publish code used for the baseline experiments. All code and additional data required for the experiments, including our splits into training and test datasets, are available on GitHub via <https://github.com/xulouzhe/CAD-SEL>.

Data Availability

The complete CAD-SEL dataset is publicly available via figshare.²⁷ CAD-SEL is distributed under the Creative Commons Attribution 4.0 International (CC BY 4.0) License, permitting use, sharing, adaptation, and redistribution in any medium, provided appropriate credit is attributed to the original authors and source.

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Author contributions

Louzhe Xu, Shurong Chen and Chunxiao Li participated in the research design, data analysis, performance of the research and the writing of the article. Jingjing Zhou, Xiaohan Shen, Jinghua Wang, Xinjue He, Tianlian Yan, Han Ma, Yizhen Zhang, Lan Li, Shu Bian, Yue Huang, Chao Lu, Yangyang Xiong, Jiang Liu, Qiang Chen, Lina Ye and Qiaona He participated in the performance of the research. Yini Ke, Youming Li and Xinli Mao participated in resources and funding acquisition. Yi Chen, Ting Li, and Chaohui Yu participated in the critical revision of the article, research design, funding acquisition, project administration, and supervision. Yi Chen had primary responsibility for final content. All authors read and approved the final manuscript.

Competing interests

The authors declare that they have no conflicts of interest regarding the publication of this manuscript.

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